

lecture 20 : examples

19/11/23

- a) find the points of max/min of $f(x,y) = xy$ over the circle $x^2 + y^2 = 1$
- b) find the points of max/min of $f(x,y,z) = y + z$ on the sphere $x^2 + y^2 + z^2 = 1$
- c) find the closest and furthest point from $(0,0)$ on the curve $2x^2 + 3y^2 - 1 = 0$

$$f(x,y) = xy \quad g(x,y) = x^2 + y^2 - 1 = 0$$

we seek $(x,y), \lambda$ s.t.
$$\begin{cases} f_x(x,y) = \lambda g_x(x,y) \\ f_y(x,y) = \lambda g_y(x,y) \\ g(x,y) = 0 \end{cases}$$

so
$$\begin{cases} y = \lambda 2x & \textcircled{1} \\ x = \lambda 2y & \textcircled{2} \\ x^2 + y^2 - 1 = 0 & \textcircled{3} \end{cases}$$

substitution?

Remember: any point you find must satisfy all three conditions

sub: $\textcircled{1}$ into $\textcircled{2}$: $x = 2\lambda(2\lambda x) = 4\lambda^2 x$

$$\Rightarrow x = 4\lambda^2 x \Rightarrow x - 4\lambda^2 x = 0 \Rightarrow x(1 - 4\lambda^2) = 0$$

$$\boxed{x=0} \text{ or } 1 - 4\lambda^2 = 0 \Rightarrow 4\lambda^2 = 1 \Rightarrow \lambda^2 = \frac{1}{4} \Rightarrow \boxed{\lambda = \pm \frac{1}{2}}$$

$$f(x, y, z) = y + z$$

$$h(x, y, z) = x^2 + y^2 + z^2 - 1 = 0$$

we seek $(x, y, z), \lambda$ s.t.

$$\begin{cases} f_x = \lambda h_x \\ f_y = \lambda h_y \\ f_z = \lambda h_z \\ h = 0 \end{cases}$$

so

find the points nearest and furthest from $(0,0)$
on the curve $2x^2 + 3y^2 - 1 = 0$
what are we optimizing?

what is the constraint?

